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Looking Beyond Nativity: The Relation of Age of Immigration, Length of Residence, and Birth Cohorts to the Risk of Onset of Psychiatric Disorders for Latinos

Margarita Alegria, PhD.[1], **William Sribney, MS**[2], **Meghan Woo, ScM**[3], **Maria Torres, MA, LMHC**[1], and **Peter Guarnaccia, PhD**[4]

1 *Center for Multicultural Mental Health Research, Cambridge Health Alliance/Harvard Medical School, Boston, MA USA*

2 *Third Way Statistics, White Lake, NY USA*

3 *Harvard School of Public Health, Boston, MA USA*

4 *Institute for Health, Health Care Policy and Aging Research, Rutgers University, New Brunswick, NJ USA*

Abstract

Past studies yield inconsistent results regarding risk of psychopathology for U.S. Latinos by nativity possibly due to differences across immigrants in their age of arrival to the U.S., their length of residence in the U.S., or birth-cohort differences. This paper seeks to document the relation of age of arrival, time in the U.S., and cohort effects on the risk of onset of psychiatric disorders using a nationally representative sample of 2554 Latinos in the coterminous United States. Risk of onset of psychiatric disorders was assessed using the World Health Organization Composite International Diagnostic Interview (WMH-CIDI, Kessler & Ustun, 2004). Findings indicate that Latino immigrants have lower risks of onset for some psychiatric disorders in their country of origin, but once in the U.S., Latino immigrants appear to experience similar risks of onset as U.S.-born Latinos of the same age. The longer Latino immigrants remain in their country of origin, the less cumulative risk of onset they experience, resulting in lower lifetime rates of disorders. These findings could potentially be due to variation in cultural and social norms and expectations across geographical contexts, differences in family structure and gender roles, as well as *artifactual*-level explanations.

INTRODUCTION

Although many studies have found that Latino immigrants have better mental health than their U.S.-born counterparts (Burnam, Hough, Karno, Escobar, & Telles, 1987; A. Ortega, Rosenheck, Alegria, & Desai, 2000; Vega, Kolody et al., 1998), researchers have begun to question the uniformity of these findings for all Latinos. In particular, cumulative evidence suggests inconsistent results regarding the risk of psychopathology by nativity (Alegría, Canino, Stinson, & Grant, 2006; Ortega & Alegría, 2002) possibly due to differences across immigrants in their age of arrival to the U.S., their length of time in the U.S., or cohort differences. Unfortunately, less is known about how to elucidate these inconsistencies.

Some researchers suggest that higher risk of illness among some Latino immigrants may result from residing in the U.S. for a longer time period (Beauvais, 1998; Rogler, Cortes, & Malgady,

Correspondence: Margarita Alegria, Ph.D. Center for Multicultural Mental Health Research 120 Beacon St. 4th floor Somerville, MA 02143. Phone: 617−503−8440 Fax: 617−503−8430 (E-mail: malegria@charesearch.org)..

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1991). Initially, the process of acculturation was thought to have positive effects on the mental health of immigrant groups (Ortiz & Arce, 1984; Rogler et al., 1991) but the more recent view is that acculturation is a gradual process whereby immigrants lose some of their culturally-defined lifestyle and adopt some of the norms and behaviors of the host country (Hunt, Schneider, & Comer, 2004; Singh & Siahpush, 2001). The acquisition of U.S. cultural values has been linked to augmented risk of health and mental health problems as a result of the disruption of family supportive networks (Rogler et al., 1991), increased intergenerational conflict, heightened family burden, and potentially weaker identification with Latino cultural values that have been associated with better mental health such as strong family ties (Finch & Vega, 2003). Yet it has become evident that such a simple causal pathway does not exist. Recent studies have looked beyond nativity to examine the effect of migration differences, such as age of arrival to the U.S. and length of U.S. residence on the mental health of Latino immigrants (J. Angel & Angel, 1992; Mills & Henretta, 2001; Portes & Rumbaut, 2001; Singh & Siahpush, 2001; Vega & Gil, 1998; Vega et al., 2002; Vega, Sribney, Aguilar-Gaxiola, & Kolody, 2004; Warheit, Vega, Khoury, & Gil, 1996). These findings provide a more complicated picture of the immigrant experience and indicate that factors beyond nativity contribute to the risk of poor mental health.

Studies conducted to date suggest that longer length of residence in the U.S. is associated with higher risk of psychiatric disorders among some Latino immigrants, particularly those of Mexican origin. Using a population survey of immigrants and U.S.-born adults of Mexican descent, Vega et al. (2004) found that 12-month rates of mood, anxiety, and substance disorders were higher for immigrants with 13 or more years of residence in the U.S. than those with less than 13 years in the U.S. (18.4% vs. 9.2%). Similarly, among Mexican Americans, longer residence in the U.S. (13 years or more) was associated with increased risk of lifetime mood disorders and substance abuse/dependence (Alderete, Vega, Kolody, & Aguilar-Gaxiola, 2000). Studies among Latino immigrant youth, particularly Cuban and Nicaraguan, have found a similar trend with positive associations found between longer time in the U.S. and past year use of cigarettes, alcohol, and illicit drugs (Gfroerer & Tan, 2003; Warheit et al., 1996). The progressive effects of acculturation and social stress have been proposed as explanations for these findings (Vega et al., 2004).

The lower prevalence of disorders for has also been attributed to the impact of length of residence on *artifactual* level explanations. For example, some symptoms may not exist in the repertoire of less acculturated immigrant Latinos while some symptom experiences more typical of less acculturated immigrants may not be included in diagnostic assessments such as “nervios”; or some symptoms may not have the same distribution or meaning across Latino immigrants residing in non-U.S. contexts (Mezzich, Jorge, & Salloum, 1994). Escobar (1998) argued that some epidemiological studies may commit a “category fallacy” by applying diagnostic assessment instruments developed in a Westernized society to Latino culture. Other investigators have suggested that somatization symptoms (rather than specific symptomatic complaints) are an expression of depression and anxiety in Latino and Asian populations (Canino, 2004; Pina & Silverman, 2004; Varela et al., 2004). In addition, the CIDI is somewhat difficult to administer in community settings with high rates of low literacy and inexperience in responding to surveys because of its complex wording (Tausig, Subedi, & Broughton, 2003).Given these factors, response biases in reports of psychopathology may be influenced by longer U.S. residence, due to greater exposure to symptom terminology used in the U.S.

Further complicating the relationship between nativity and the risk for psychiatric illness is how the observed patterns may vary by the age of arrival and developmental stage of the immigrant to the U.S. These factors may signal critical windows of opportunity for interfacing with U.S. society, and consequently differential risks for negative outcomes. Research findings suggest that Latino immigrants are at higher risk for psychiatric disorders when immigrating

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during two life cycle periods: before the age of 16 (Vega et al., 2002; Vega et al., 2004), or after the age of 35 (Mills & Henretta, 2001).

Adolescence is a time when the formation of a stable identity and sense of group membership occurs (Herman, 2004). However due to social status norms around race and experiences of discrimination in the U.S., minority adolescents often develop identities as low-status individuals which can make deriving a positive self-identity more difficult (Erickson, 1968). For those immigrating at younger ages, this process may be exacerbated as they may be at higher risk for “not fitting in”, experiencing peer rejection by their U.S. born peers, and hence isolation and loneliness (Diversi & Mecham, 2005). These early age immigrants, may further confront a challenging socialization process, fraught with intergenerational conflict of parents trying to hold on to the values and norms of their native land while the children experience pressure to socialize into U.S. culture (Louie, 2003; Suarez-Orozco & Todorova, 2003). This conflict is more likely to occur among immigrants arriving to the U.S. as young children as they are more likely to distance themselves from the protective effect of Latino family and culture as a result of the natural developmental process whereby children begin to assert their independence from their family unit (Elder, Broyles, Brennan, Zuniga de Nuncio, & Nader, 2005) and place greater dependence on their peers. This natural assertion of independence often occurs as Latino children are required to attend U.S.-based schools, resulting in greater exposure to the English language and U.S. culture from their teachers, peers, and curriculum. Therefore, due to larger societal forces, those who arrive in the U.S. at an early age may have greater integration into U.S. culture and potentially weaker identification with Latino cultural values that have been associated with better mental health (Alegría et al., in press; Finch & Vega, 2003).

For those immigrating at a younger age, researchers (Vega et al., 2002) have found that illicit drug use was more likely to be initiated if the immigrant arrived in the U.S. before 24 years of age. They also found that a young age (0−16 years) of arrival into the U.S. was associated with increased likelihood of a 12-month diagnosis of mood, anxiety, and substance disorders (Vega et al., 2004). Earlier arrival and longer length of residence in the U.S. as well as developmental stage at age of arrival are believed to lead to increased socialization into U.S. values and lifestyle and to exposure to more problem behaviors for minority youth (Harker, 2001).

Those immigrating at age 16 or older may be further along in their process of identity development, thus making them less vulnerable to challenging socialization processes such as being accepted or rejected by school peers. In addition, intergenerational conflict is minimized as these individuals have spent their most formidable period of identity development in their country of origin. As a result, later arrival immigrants tend to share norms and values closer to their parents due to greater identification with Latino cultural values. Instead, their attention might be more directed at social mobility, without having to deal with getting integrated into Americanized social networks, mainly seeking better job opportunities than those in their native land.

However, immigrants who came to the U.S. at or after age 35 may have even less opportunity for exposure to U.S. society, allowing for better retention of their native language and culture norms (Elder et al., 2005). However, they may also be at increasing risk of feeling uprooted, of not achieving English linguistic proficiency, or of possibly not transporting their educational or professional qualifications to the host society (Kaplan & Marks, 1990), thereby increasing their risk of mental disorders. For example, Mills and Henretta (Mills & Henretta, 2001) found that immigrants who came to the U.S. after age 35 were more depressed than those arriving at an earlier age. Other studies have indicated that difficulties associated with immigration late in life, such as low English language acquisition and social isolation, may undermine an older person's morale, leading to augmented risk for illness and dysfunction (Angel & Angel,

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1992; Angel, Angel, Geum- Yong, & Markides, 1999).In addition, isolation from the mainstream culture may also prevent older immigrants from serving as protective agents and mediators for their children's interactions with harmful aspects of U.S. culture (Berry, 1998; Elder et al., 2005). This may and actually lead to role reversals, where their children take a more authoritative role and erode the affiliative obedience and respect for the adults (Zhou, 1997).

Also of relevance for the risk of psychiatric disorders might be cohort differences. Several studies suggest that prevalence for both depression and substance use has increased and that age of onset has decreased for successive birth cohorts (Hasin & Link, 1988; Lewinsohn, Rohde, Seeley, & Fischer, 1993). It is thought that large societal changes that occurred in the post-WWII era such as feminism, an increased tolerance for drug use, and a change in the structure of the American family drastically affected rates of depression and substance-use disorders (Grant, 1996; Hasin & Link, 1988; Warner, Kessler, Hughes, Anthony, & Nelson, 1995). Joyce and colleagues (Joyce, Oakley-Browne, Wells, Bushnell, & Hornblow, 1990) found that in persons born after 1960, the six-month prevalence of psychiatric disorders was greater than the lifetime prevalence in the oldest cohort (born in 1921−1930). Other researchers report that depressive and drug use disorders are more common in later-born cohorts (born after 1960) compared to earlier cohorts (before 1960, (Holdcraft & Iacono, 2003). How these factors affect Latino immigrants is less clear as there may be other events that are critical markers of the preand post-1960 period such as the Cuban Revolution for Cubans (the overthrow of Batista's regime in 1959 leading to a social revolution that adopted Marxist principles and the nationalization of foreign-owned property) and “Operation Bootstrap” for Puerto Ricans (an ambitious program to change Puerto Rico from an agricultural society to an industrial working class society by the 1960s).

Wide variations in sample population, study design and methodology, assessment tools, and measurement in the aforementioned studies makes it difficult to gain a complete picture of the impact of age of arrival, length of U.S. residence, and cohort effects on the risk of psychiatric illness among Latinos in the U.S. This paper seeks to address some of these limitations and build upon past work by presenting data from the National Latino and Asian American Study (NLAAS), a national epidemiological and service use study of Latinos and Asian Americans. Data from the NLAAS provide the opportunity to explore the relation of age of arrival, length of U.S. residence, and cohort effects on the risk of onset of psychiatric disorders for Latinos among the major Latino groups in the U.S.

METHODS

Sample

The NLAAS includes large numbers of respondents from Latino ethnic subgroups. As described in detail elsewhere (Heeringa et al., 2004), the NLAAS is a nationally representative survey of English- and Spanish-speaking household residents ages 18 and older in the non-institutionalized population of the coterminous United States. Latinos were divided into four subgroups: Mexican, Puerto Rican, Cuban, and Other Latino. The final sample comprised 2,554 Latinos with a response rate of 75.5%. This includes an NLAAS Core sample, designed to provide a nationally-representative sample of all Latino origin groups regardless of geographic residential patterns, and NLAAS-HD supplements, designed to oversample geographic areas with moderate to high density (≥5%) of targeted Latino households in the U.S. Weighting reflecting the joint probability of selection from the pooled Core and HD samples provides sample-based coverage of the full national Latino population. The NLAAS weighted sample is similar to the 2000 Census in sex, age, education, marital status, and geographical distribution (data not shown) but different in nativity and household income, with more U.S. immigrants and lower income respondents in the NLAAS sample. This discrepancy

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may be due to Census undercounting of immigrants (Anderson & Fienberg, 1999; United States General Accounting Office, 1998) and non-inclusion of undocumented workers (Margolis, 1995).

Data Collection

Data were collected by the Institute for Social Research at the University of Michigan between May 2002 and November 2003. Eligibility criteria for the Latino sample of the NLAAS included age (18 years or older), ethnicity (Latino, Hispanic, or Spanish descent), and language (English or Spanish). Two hundred seventy-five certified bilingual Latino interviewers obtained written informed consent and administered the NLAAS interview in Spanish or English. Approximately half of the participants were monolingual Spanish speakers or had limited English proficiency and requested the interview in Spanish. For quality control, a 15% random sample of each interviewer's completed interviews was re-contacted for validation.

The Institutional Review Board Committees of the Cambridge Health Alliance, the University of Washington, and the University of Michigan approved all recruitment, consent, and interviewing procedures. A detailed description of the NLAAS data collection procedures are described elsewhere (Pennell et al., 2004).

Measures

Interviews were designed to be linguistically and culturally appropriate for each Latino subgroup. The key variables and scales are more fully described in Alegria et al., (2004). (***Sociodemographic and immigration measures*** included in the analysis are **gender; age; nativity** (*U.S. born, immigrant*); **immigrant age of arrival**, the age at which immigrants arrived in the U.S. (*0−6, 7−17, ≥18*); **birth cohort** (*born before or after January 1, 1960*); and **immigrant length of residence in the U.S.** (in U.S. 5 years or less, in U.S. for more than 5 years). The **in country of origin** category examines retrospectively the risk experienced by immigrants in their country of origin before arrival to the U.S.

Diagnostic measures—for lifetime and last 12-month prevalence of psychiatric disorders were assessed using the diagnostic interview of the World Mental Health Survey Initiative version of the World Health Organization Composite International Diagnostic Interview (WMH-CIDI, Kessler & Ustun, 2004), based on criteria of the Diagnostic and Statistics Manual of Mental Disorders, version 4 (DSM-IV, American Psychiatric Association, 1994). Table 1 lists the lifetime and 12-month composite diagnostic categories used in this analysis: **any depressive disorder** (dysthymia, major depressive episode); **any anxiety disorder** (agoraphobia, social phobia, generalized anxiety, post traumatic stress, panic); and **any substance-use disorder** (drug abuse, drug dependence, alcohol abuse, alcohol dependence). Diagnoses were made with DSM-IV organic exclusion rules. The validity of the earlier CIDI diagnostic assessments were consistent with those obtained independently by trained clinical interviewers (Wittchen, 1994). Findings of the instrument show good concordance between DSM-IV diagnoses based on the WMH-CIDI and the SCID (First, Spitzer, Gibbon, & Williams, 1998) for mood and substance disorders. Ages of onset for these classes of disorder were determined by retrospective recall based on separate probes: age of onset of the disorder/symptoms. Onsets for immigrants are further categorized in Table 2 according to whether the onset (based on the reported age when respondent first experienced the symptoms and substantial impairment) was reported to have occurred before or after their arrival in the U.S. (based on the age when the respondent arrived to the U.S.). We compare immigrants who arrived between the ages of 0−6, 7−17 and ≥18 years. This grouping is consistent with previous literature (Capps, Fix, Ost, Reardon-Anderson, & Passel, 2004).

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Statistical Analyses

Probability sampling weights were modified using a square-root transformation to reduce variance of the weights and estimates based on them. After this transformation, weights were adjusted using a ranking procedure (Little & Rubin, 2002) to the Census 2000 age by gender distribution of Latinos (U.S. Census Bureau, 2000), while maintaining the observed sample proportions of Latino sub-ethnicities. Table 1 presents weighted sample and subsample frequencies. Significance tests for Table 1 were performed using a Rao–Scott statistic that is the Pearson χ^2 test for contingency tables adjusted for the survey design (Rao & Scott, 1984; Rao & Thomas, 1989). Incidence rates shown in Table 2 are crude (not age-adjusted) weighted estimates based on recall of age of onset and person-years of retrospectively observed time at risk (i.e., years observed before any onset for those with disorders and years prior to the date of interview for those without disorders). The relative ordering of time of onset and arrival in the U.S. could not be determined for those subjects whose onset and arrival occurred during the same year because the exact month of onset of the disorder as well as the month of arrival into the U.S. were not assessed in the NLAAS survey. Onsets that were reported during the same year as arrival in the U.S. were included in the "before arrival in the U.S." category.

In this study, we use Cox proportional hazards models to evaluate whether the risk of onset for depressive, anxiety, and substance-use disorders differs between immigrants and U.S.-born. Men and women were analyzed separately since previous studies have indicated differential rates and different age patterns of prevalence in this population, especially for depressive and substance-use disorders (Kasen, Cohen, Chen, & Castille, 2003). Because we hypothesize that early integration and socialization into U.S. culture would eliminate or moderate the protective effects of Latino culture, three groups were chosen for the hazards models: U.S.-born, immigrants who were 0−6 years of age at time of arrival into the U.S., and immigrants who were seven years of age or older at time of arrival. Table 3 gives weighted Cox proportional hazards models that estimate the risk of onset as a function of age, based on a synthetic cohort that compares persons retrospectively at the same age in the past. To partially control for the fact that this retrospective analysis matches persons from different birth cohorts, we included terms for decade of birth in our models. The final models we present only retained a single term for birth cohort (*born before 1960*) since this was statistically adequate to fit the observed birth-year effect. The Cox proportional hazards assumption was assessed using tests of Schoenfeld residuals (Schoenfeld, 1982) and by visual examination of loglog plots of the estimated survival curves. Standard error estimates from the Cox proportional hazards models were adjusted for the sampling design through a first-order Taylor series approximation (Binder, 1992; Lin, 2000; Lin & Wei, 1989) and design-based Wald tests (Korn & Graubard, 1990) were used to test significance.

Figures 1-4 show estimated hazards from weighted exponential survival models in which constant hazards were fit for ages 0−10, 11−15, 16−20, 21−25, 26−30, 31−40, and 41−50 years for each of three groups: U.S.-born, immigrants whose age of arrival was 0−6 years, immigrants whose age of arrival was 7 years or older. The term *born before 1960* was also included in these models. In addition, to model the effect of length of residence in the U.S. for immigrants whose age of arrival was 7 years or older, terms of x and x^{-2}, where x is years in the U.S., were included in the exponential models. These powers of x were chosen after automatically examining several two-term fractional polynomials using Stata's fracpoly command (Royston & Altman, 2004).The term x represents a linear trend over time and x^{-2} represents a trend that goes to zero after some time (i.e., no effect of years in the U.S. after some time). Although x and x^{-2} gave the best fit for the effect of length of residence, other combinations of powers of x gave similar results; thus, the estimation of the effect of length of residence in the U.S. was not highly dependent on the powers of x chosen. Before plotting the hazards in Figures 1-4, estimates were slightly smoothed to make visual examination easier using a robust nonlinear

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smoother (Velleman, 1980), which smoothed a maximum span of 4 years at a time. All analyses were conducted using the Stata statistical software package, version 8.2 (StataCorp, 2004).

RESULTS

Table 1 shows the characteristics of the Latino sample of the NLAAS. The rightmost column in Table 1 gives statistical tests of the differences among the four subgroups shown: U.S. born, immigrants whose age of arrival is 0−6 years, immigrants whose age of arrival is 7−17 years, and immigrants whose age of arrival is ≥18 years. The majority of the sample are immigrants (58%), and of these, well over half (60%) immigrated to the U.S. at age 18 or older. The weighted sample is evenly split between male and female respondents (51.5% vs. 48.5%). The population is quite young with roughly half of Latinos less than 35 years of age and more than two thirds born after the year 1960. The vast majority of immigrants have resided in the U.S. for more than 5 years and almost 80% of the immigrants whose age of arrival was 18 or older have been in the U.S. for more than 5 years.

Lifetime and 12-month prevalence rates by class of disorder are also shown in Table 1. Overall, any lifetime depressive disorder was the most prevalent (15.6%) followed by any lifetime anxiety disorder (15.4%) and any lifetime substance-use disorder (11.3%). For 12-month disorders, 9.6% of respondents reported any 12-month anxiety disorder, 8.9% reporting any 12-month depressive disorder, and 2.9% reporting any 12-month substance disorder. U.S.-born Latinos and immigrants whose ages of arrival were age 0−6 years had higher rates for every disorder category than the immigrants who arrived after age 6. The U.S. born and early childhood arrival immigrants had very similar lifetime and 12-month rates of depressive and anxiety disorders. For substance-use disorders, the early childhood arrival immigrants had rates that were roughly half that of the U.S. born. Of the three classes of disorders, only prevalence rates of substance-use disorders (both lifetime and 12-month) were significantly different among the nativity/age of arrival groups.

Table 2 reports the number of onsets observed, and crude (not age-adjusted) incidence rates by nativity and age of arrival to the U.S. Incidence rates for immigrants are categorized according to whether the onset occurred before or after their arrival in the U.S. Incidence rates for depressive disorders for the U.S. born and immigrants *after arrival in the U.S.* were similar, ranging from 4.2 per 1000 person-years for immigrants in the U.S. whose age of arrival was 18 years or older to 5.3 per 1000 person-years for immigrants whose age of arrival was 0−6 years.

The incidence rate for anxiety disorders observed for U.S. born (5.2 per 1000 person-years) was similar to that for immigrants in the U.S. whose age of arrival was 0−6 years (5.8 per 1000 person-years). Incidence rates after arrival in the U.S. for immigrants arriving at age 7−17 and ≥18 were lower (2.6 and 2.8 per 1000 person-years, respectively). This is not surprising since anxiety disorders are typically of early onset and these immigrants spent much or all of their early years in their country of origin.

In contrast to the other disorders, incidence rates for substance-use disorders were very low for immigrants in their country of origin before arrival in the U.S. Once in the U.S., later arrival (age 7−17 and ≥18) immigrants had lower incidence rates than the U.S. born, with early arrival (age 0−6) immigrants having rates about halfway in between. Direct comparison of incidence rates for the immigrant subgroups before and after arrival in the U.S. is difficult since one must fully control for age both before and after arrival in the U.S. Differences are best examined using the Cox models shown in Table 3, where age of arrival is included as a time-varying covariate.

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Table 3 shows Cox proportional hazards models contrasting the risk of onset of depressive, anxiety, and substance-use disorders by nativity and age of arrival to the U.S. Models were fit separately for males and females since the proportional hazards assumption was violated in models that combined both genders. Ages of onset were retrospectively estimated using CIDI probes for onset of symptoms and impairment. Table 3 shows results for three groups: U.S.-born, immigrants arriving between the ages of 0−6, and immigrants arriving at age 7 or older. The group of immigrants arriving at age 7 or older had three hazards modeled: the hazard experienced in their country of origin before arrival to the U.S., the hazard experienced during their first 5 years in the U.S., and the hazard experienced after 5 years in the U.S. An additional term for all persons (birth before or after January 1, 1960) was included in the model to partially control for year of birth.

We found no significant difference in risk of disorders between U.S.-born Latinos and Latino immigrants arriving in the U.S. between the ages of 0−6. Overall, immigrants arriving at age 7 or older appeared to experience lower risk for disorders at all time periods in their lives compared to the U.S. born, although only a few differences were significant. For depressive disorders, male immigrants arriving to the U.S. at age 7 or older experience lower risk in their country of origin before arrival to the U.S. compared to U.S.-born males (HR = 0.56). Among women, those arriving to the U.S. at age 7 or later experienced lower risk for depressive disorders during their first 5 years in U.S. compared to U.S.-born females (HR = .54). For anxiety disorders, both male (HR = 0.55) and female (HR = 0.65) immigrants who arrive to the U.S. at age 7 or older, experience significantly lower risk of any anxiety disorder when still in their country of origin compared to their U.S.-born counterparts. Table 3 also demonstrates that all male and female immigrants arriving to the U.S. at age 7 or older are at significantly lower risk of any substance-use disorder compared to the U.S. born, both in their country of origin and after arrival in the U.S.

We found that birth cohort impacts the risk for any depressive disorder in both genders and substance-use disorder in females. Males (HR = 0.46) and females (HR = 0.36) born before 1960 were at significantly lower risk for any depressive disorder compared to those born after 1960. Women born before 1960 were also at significantly lower risk for any substance use disorder compared to women born after 1960 (HR = 0.41).

The five-year division in Table 3 for looking at length of residence was chosen arbitrarily; results were similar using slightly different cutoffs, and using multiple categories for length of residence was problematic because too few onsets fell into each category to render comparisons statistically meaningful. Given this limited statistical power, effects of length of residence are best modeled using a continuous measure. We present results of such a modeling procedure graphically in Figures 1-4. We fit piecewise exponential survival models (i.e., piecewise constant hazards) for each of the six sex by disorder groupings as in Table 3. Constant hazards were fit for 7 age categories (ages 0−10, 11−15, 16−20, 21−25, 26−30, 31−40, and 41−50) for each of three groups: U.S. born, immigrants whose age of arrival was 0−6 years, and immigrants whose age of arrival was 7 years or older. Two additional terms were added for immigrants whose age of arrival was 7 years or older to model the trend of length of residence in the U.S.: one term was the linear time in the U.S. and the inverse of the square of time in the U.S. (see Methods for details).

In order to see length of U.S. residence effects for immigrants with later ages of arrival, we looked at immigrants at precise ages of arrival; otherwise, any short-term effects would get averaged with longer-term effects. Figures 1-4 show hazard estimates for immigrants at ages of arrival 12, 18, and 25. Note that although only these particular ages are shown, the estimates are based on models fit using all immigrants. Hence, the time in the U.S. effects shown must be considered a representative “average” of all later arrival immigrants. Indeed, this is exactly

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how the figures were produced: a representative time-in-the-U.S. effect is shown overlaid on the general age trend for later arrival immigrants.

Figures 1-4 show the estimated risk for the onset of any depressive and substance use disorders separately for males and females. A similar analysis was performed for risk of onset of any anxiety disorder; however these data are not shown. In general, immigrants who arrive in early childhood (age 0−6 years) appear to be protected against risk of onset during their first few years in the U.S. After this period, however, for depressive and anxiety disorders, their risk of onset rapidly rises. In other words, onsets of depressive and anxiety disorders for early childhood immigrants seem to be merely delayed compared to the U.S.-born. The pattern of risk for substance-use disorders is similar; with a short period of protection after arrival followed by an increase in risk. However, the level of risk never reaches the highest levels experienced by U.S.-born Latinos.

The figures also demonstrate that immigrants who arrived in the U.S. after early childhood (≥7 years of age) experience significantly lower risks of depressive, anxiety, and substance-use disorders in their country of origin compared to U.S.-born Latinos of the same age and sex. After arrival in the U.S., their risk of onset is low for a year or two, and then begins to increase, eventually reaching levels similar to that of the U.S.-born and early-arrival immigrants of the same age. The only exception to this trend is substance-use disorders in later-arrival female immigrants, whose risk remains near zero their entire lives. The increase in risk after arrival in the U.S. is quite dramatic for depressive disorders in male early-arrival immigrants and substance-use disorders in male early-arrival immigrants. Risk for onset of depressive disorders in later-arrival immigrants has another interesting pattern: after roughly age 30 in males and females, the risk of onset appears to increase as they age into their late thirties and into their forties, in contrast to the U.S.-born and early childhood arrival immigrants, whose risk continues to decline at these ages. However, since fewer onsets were observed at these later ages among all persons, there was insufficient statistical power to adequately test the significance of this observation.

Discussion

In this study, we found significant differences in the patterns of onset of psychiatric illness among Latinos in the U.S. by nativity and by age of arrival among immigrants. However, there are several limitations to this study. The Cox models and statistical modeling used to produce Figures 1-4 are based on retrospective recall of the age of onset of a disorder. It is possible that recall bias may have led to an underestimation of hazard rates in the country of origin. These models assume the same conditional risk for onset in a given year of life among Latinos who vary in age of interview. This seems an unrealistic assumption given the significant differences in lifetime prevalence rates across age cohorts (Kessler, Berglund, Demler, Jin, & Walters, 2005).We attempted to control for this by including a birth-cohort term in our model; however, this may have been insufficient to control for possibly complex cohort effects. The statistical modeling used to produce Figures 1-4 fit the trend for time in the U.S. for later arrival immigrants under the assumption that this trend was the same for all immigrants arriving after age 6. The trend for time in the U.S. shown in Figures 1-4 (which is overlaid on the general age trend) should be considered the “average” effect for all immigrants with later ages of arrival. Ideally, one would look at time in the U.S. effects for two or more categories of immigrants with later ages of arrival (e.g., 7−17 and ≥18), but there was not sufficient sample size to do so. Therefore, these results should be replicated with a larger data set that allows for a more sensitive examination of risk of onset in country of origin compared to risk of onset in the U.S. with ideally a longitudinal design.

Our findings suggest that later age of arrival (at 7 years of age or older) into the U.S. might be linked to later onset of psychiatric disorders for depressive (except for immigrant females),

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and substance-use disorders. Our results indicate that during the time Latinos reside in their country of origin prior to immigration, they are protected against the onset of psychiatric disorders. Once they arrive in the U.S., the risk of onset for these later-arrival immigrant Latinos approaches to that of U.S.-born Latinos of the same age. Hence, the longer they remain in their country of origin, the less cumulative risk of onset of the disorder they experience. Then once in the U.S., these later-arrival immigrants never experience high risks of onset, which combined with their earlier period of low risk, results in lower lifetime prevalence rates. This is especially true if they remain in their country of origin during ages at which one sees the greatest risk of onset for the disorder (teens and early twenties for depressive and substance-use disorders and childhood and early teens for anxiety disorders). This is in contrast to early arrival immigrants who experience very high risks of onset shortly after arrival in the U.S. and quickly "catch up" to the U.S. born in levels of cumulative risk, and hence have similar lifetime prevalence rates.

Our results indicate that all of the protective effect experienced by immigrants appears to occur in their country of origin; there appears to be no protective effect for risk of onset for immigrant Latinos after arrival in the U.S. compared to U.S.-born Latinos of the same age (who have not yet experienced onset). Our findings (Alegría et al., under review) in a complimentary paper suggest that after we control for positive family ties and self-perception of good social position, Latino immigrants in the U.S. show *increased* 12-month rates of depressive and anxiety disorders as compared to U.S.-born Latinos, while for substance-use disorders, religious values and neighborhood context appear more relevant determinants of 12-month disorder rates. These findings for 12-month rates support the findings in this paper from the analyses of onset that once immigrant Latinos arrive to the U.S., they have no special protective factors compared to U.S.-born Latinos of the same age.

Protective patterns of familism (sense of familial obligation to provide material and emotional support; (Marin & Vanoss-Marin B, 1991); and affiliative obedience (where children are raised with strong values towards parental respect and deference to elders) (Harker, 2001) may be hard to retain after the immigrants leave their native country. Affiliative obedience and familism may rapidly deteriorate after arrival in the U.S., where childrearing norms consistent with U.S. values likely promote individualism and independence from parents rather then familism and strong affiliative obedience. There is some evidence that these cultural values (affiliative obedience and familism) are related to living structure in extended families, with greater support from social networks, and relationships of reciprocal obligation that might serve to buffer stressful life circumstances linked to an increased risk of psychiatric disorders. These cultural values and practices may facilitate a sense of control and self-efficacy conducive to better mental health (Escobar, 1998). Hence, once living structure is disrupted by immigration to the U.S. (e.g., leaving behind extended family members and the social networks of their country origin), these benefits may be quickly lost.

Intergenerational conflict that might arise due to tensions when children and parents do not share the same values, norms, lifestyle, and role expectations also may arise soon after arrival in the U.S. Socialization into U.S. society may weaken protective resources of Latino culture such as family support and cohesiveness, and increase family conflict due to different rates of cultural change between parents and children (Portes & Rumbaut, 2001; Vega & Gil, 1998; Vega et al., 2004). At the same time, socialization in U.S. society might imply the rejection of these norms and values that ensure active social support and the dilution of the perception of control and self-efficacy gained by relationships of reciprocal obligation. The potential consequence might be a detrimental loss of belonging and social isolation as well as increased intergenerational conflict.

As previously mentioned, the lower lifetime prevalence of disorders for immigrants living longer in Latin American countries could also potentially be due to *artifactual* level

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explanations. It might be that construct bias, defined as the incomplete overlap of the psychiatric illness construct across less acculturated Latinos could account for some of these differences. One important finding from our initial analyses (Alegría et al., under review) was that for most disorders, Puerto Ricans, who are U.S. citizens and are therefore constantly exposed to U.S. terminology, exhibit no protective effect of nativity, while Mexicans demonstrate a protective effect of nativity. One possibility is that Mexican immigrants born in Mexico may have different understandings of symptom assessment questions and therefore may endorse psychiatric disorder screening probes differently than their U.S. born counterparts. In the NLAAS study population, many respondents have recently emigrated from Latin America, possibly influencing their understanding of and response to the diagnostic criteria of the NLAAS instrument.

The finding for depressive disorders that there is no difference in the risk of onset between immigrant Latina women in their country of origin and the U.S. born is not necessarily contradictory to the possibility of artifactual level explanations. It may only imply that the risks are even higher for less acculturated immigrant women, but given artifactual explanations, their risk appears similar to U.S.-born Latina women. Additional work on how the symptom probes work for different immigrant groups is currently underway to test this possible explanation. The risk for depressive disorder might be high for immigrant women in their country of origin because domestic violence and oppression might be as dominant in their countries of origin as in the U.S. In a comparative study of violence against pregnant Mexican women, overall prevalence of violence did not differ significantly between Mexican women in Morelos, Mexico and Mexican women in Los Angeles County, California (Castro, Peek-Asa, Garcia, Ruiz, & Kraus, 2003). However, other studies have found significant differences in perception and reporting of domestic abuse finding that Anglo-American women perceive more types of behavior as being abusive and are less tolerant towards domestic violence than Mexican-American women (Torres, 1991). Similarly, Garcia et al. (Garcia, Hurwitz, & Kraus, 2005) found that highly and moderately acculturated Latinas were more likely to report intimate partner violence compared to the least acculturated Latinas. These findings suggest that rates of domestic violence may be even higher among less acculturated Latinas and those still residing in their country of origin, potentially placing them at higher risk for depression. Domestic violence has been consistently shown to place women at augmented risk for depressive disorders (Golding, 1999; Romito, Turan, & De Marchi, 2005).

The Cox proportional hazards model shown in Table 3 assumes proportional hazards throughout the entire time period; the piecewise exponential models used to produce Figure 1-4 do not need to make such proportional hazards assumption, and closely fits the observed hazard, whatever its pattern. There is, however, a general overall consistency between the results of Table 3 and Figures 1-4 across the genders and disorders, which is reassuring and makes it highly unlikely that the effects that we describe are artifacts of the statistical procedures used. Our findings reveal the potential nonlinearity of the impact of length of U.S. residence on risk of onset of psychiatric disorder, suggesting that models assuming a linear risk might fail to identify important patterns. The evidence thus far suggests the importance of age of arrival for the level of cumulative risk of onset of disorder observed at different time periods.

The results presented in Figures 1-4 suggest that, overall, Latino immigrants appear protected from onset of disorder in their country of origin and immediately upon arrival in the U.S. (the only exception being depressive disorders in Latina immigrants as previously noted). However, a few years after arrival, risk of onset for both early and late arrival immigrants reaches the level of the U.S. born. Previous research (Turner & Lloyd, 2003, 2004) has linked stress burden, or a high level of cumulative exposure to major stressors (such as those associated with poverty, work instability, discrimination), with increased risk for the occurrence of depressive, anxiety,

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and substance-use disorders. In Latin American countries, it is common for children to maintain strong bonds with their parents and extended family well into adulthood. Latino children often continue to live under the same roof and remain financially dependent on their parents until marriage. Extended exposure to this protective environment for Latinos in their country of origin may result in the lowered cumulative risk for psychiatric disorders seen in our data. Not only is the cumulative risk lower, but the highest period of risk experienced is shifted to later ages relative to the U.S. born, who may have weaker familial bonds, and hence experience social stressors linked to being independent and having financial and social role responsibilities at an earlier age.

Risk for onset of depressive disorders in later-arrival immigrants has another interesting pattern: after roughly age 30 in males and females, the risk of onset appears to increase as they age into their late thirties and into their forties, in contrast to the U.S. born and early childhood arrival immigrants, whose risk continues to decline at these ages. There are several potential explanations for these findings. Later-arrival immigrants might not benefit from the same level of achievement as their younger counterparts nor reap the same rewards of their hard work. Also, later-arrival immigrants may become increasingly aware of ethnic and racial categories used by the U.S. society that seem unfair and pejorative, particularly when they have an already ingrained identity coming from a majority culture. This may increase the risk of psychopathology, as they may experience many years of living in the U.S., while not fully integrated into U.S. society.

Another potential explanation is that during the late thirties and early forties, family structures change as a result of children leaving the household, with potentially stronger impacts for closely knit later-arrival Latino families (Borland, 1982; Bremer & Ragan, 1977). These family changes might be experienced as isolation and decreased family supports by later-arrival immigrants. This isolation may be intensified for older arrival immigrants due to general difficulties with language acquisition among adults. When children, who have gained greater English language proficiency through school systems, leave the household, older immigrants who have not yet mastered the English language may also become linguistically isolated. For the U.S. born and early arrival Latino immigrants, these family changes might be welcomed, seen as liberation of parental burdens and as opportunities to engage in outside interests. A related explanation is that as later-arrival immigrants start their middle age with the expectation of continued family affiliation, with the expectation that their children will take care of their elders. However, these expectations might not materialize in the host society, leading to regret and disappointment in later arrival immigrants (Foner, 1997)

Figures 1-4 show a slightly different pattern for onset of substance-use disorders than for onset of other disorders. Later-arrival male immigrants experience increased risk of onset for substance-use disorders after arrival in the U.S., peaking around age 27, but the peak hazard that they experience is only about one third of what U.S.-born and early arrival immigrant experience in their late teens; hence, the later-arrival immigrants never reach the same levels of the U.S. born in their cumulative risk for substance-use disorders. This suggests that socialization about alcohol and drug use in Latino countries and within Latino families exerts strict social norms against substance use that are imprinted before coming to the U.S. For female immigrants, the differences in rates between early and later arrival female immigrants or the U.S. born is so great that protective factors, possibly family cohesion (Maton, 1993), and strong normative behaviors linked to Latina gender roles (e.g., early childrearing and family responsibilities) may explain the difference in rates. This is particularly relevant given the universal drop in risk of substance use after age 30 for Latina females. The pattern of risk of onset of substance disorders for later arrival Latino males and females and its relationship to length of residence in the U.S, differs from other research conducted with Cuban Americans (Turner & Gil, 2002) and Mexican Americans (Vega, Alderete, Kolody, & Aguilar-Gaxiola,

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1998) that found length of U.S. residence to be an important predictor of risk. However, these studies failed to simultaneously model both age of arrival and length of U.S. residence, and their results were likely indicative of the fact that immigrants with longer lengths of U.S. residence were more likely to have earlier ages of arrival.

Consistent with previous studies (Hasin & Link, 1988; Holdcraft & Iacono, 2003; Joyce et al., 1990; Lewinsohn et al., 1993) our findings also demonstrate an impact of birth-year cohort on risk for psychiatric illness. Many hypotheses have been proposed to explain why cohorts born after 1960 show increased risk of depressive disorders among men and women and risk of substance use disorders among women. Most commonly, researchers have proposed that the escalation in rates among younger cohorts results from dramatic societal changes and shifts in the family structure which occurred after World War II and the women's movement of the 1960s and 1970s (Kasen et al., 2003; Kessler & Walters, 1998). Since this time, younger cohorts have been exposed to increasingly higher rates of parental divorce, single parent households, maternal employment, and teenage childbearing compared to those born before 1960, which may contribute to their higher rates of depression and substance use (Kasen et al., 2003; Kessler & Walters, 1998).Younger cohorts of women have been found to be particularly vulnerable to higher rates of depression and substance use, possibly due to the stress of juggling multiple roles (Kasen et al., 2003). Regarding substance use and cohort effects, increased rates of substance use among younger generations may be attributed to changes in attitude towards alcohol and drug use that occurred during the 1960s and 1970s. In particular, changes resulting in the lower stigmatization of women's drinking and drug use over the past few decades may account for the cohort effect presented in our findings. This could also be linked to particular issues in migration patterns, with greater exposure to discrimination as Latino immigration in general increases in intensity after 1960 for all the major groups.

Differences in risk for psychiatric disorder by cohort have also been attributed to selective recall of memory or changes in labeling of symptoms over time. It has been proposed that the cohort effect does not in fact exist (Giuffra & Risch, 1994; Simon & VonKorff, 1992, 1995). Rather, it is believed that recall failure increases with age and over time resulting in higher recall failure among older cohorts and thus the false impression of a cohort effect (Kasen et al., 2003; Weissman, Leaf, Holzer, Myers, & Tischler, 1984).Alternately, changes in social norms, stigmatization, and labeling of psychiatric disorders in recent decades may also lead to the observed increase (Weissman et al., 1984).Since later-arrival immigrants are, in general, older, and since the high risk period for onset would have occurred, in general, some years in the past in their country of origin, recall failure would tend to bias downward the estimates for hazards for immigrants in their country of origin. By including a term for year of birth before or after 1960 in our models in Table 3 and Figures 1-4, we attempted to control for this possible bias.

These results point to the importance of conducting longitudinal and more in-depth qualitative studies that might allow us to better understand the differential risk by age of arrival in the U.S. and birth cohorts in Latino groups. These findings also suggest that it is necessary to view risk for psychiatric disorders within a developmental context trying to see how different environments protect or reduce the risk for psychiatric disorders, particularly during certain age periods. Interpretations of symptomatic manifestations in different contexts, depending on age, might also matter, indicative that the social construction of illness might be integrated differently depending on where you live. More work to test artifactual explanations of these differences as well as comparisons of epidemiological data collected in Latino countries as part of the World Health Initiative might help us elucidate these different patterns of risk and resiliency.

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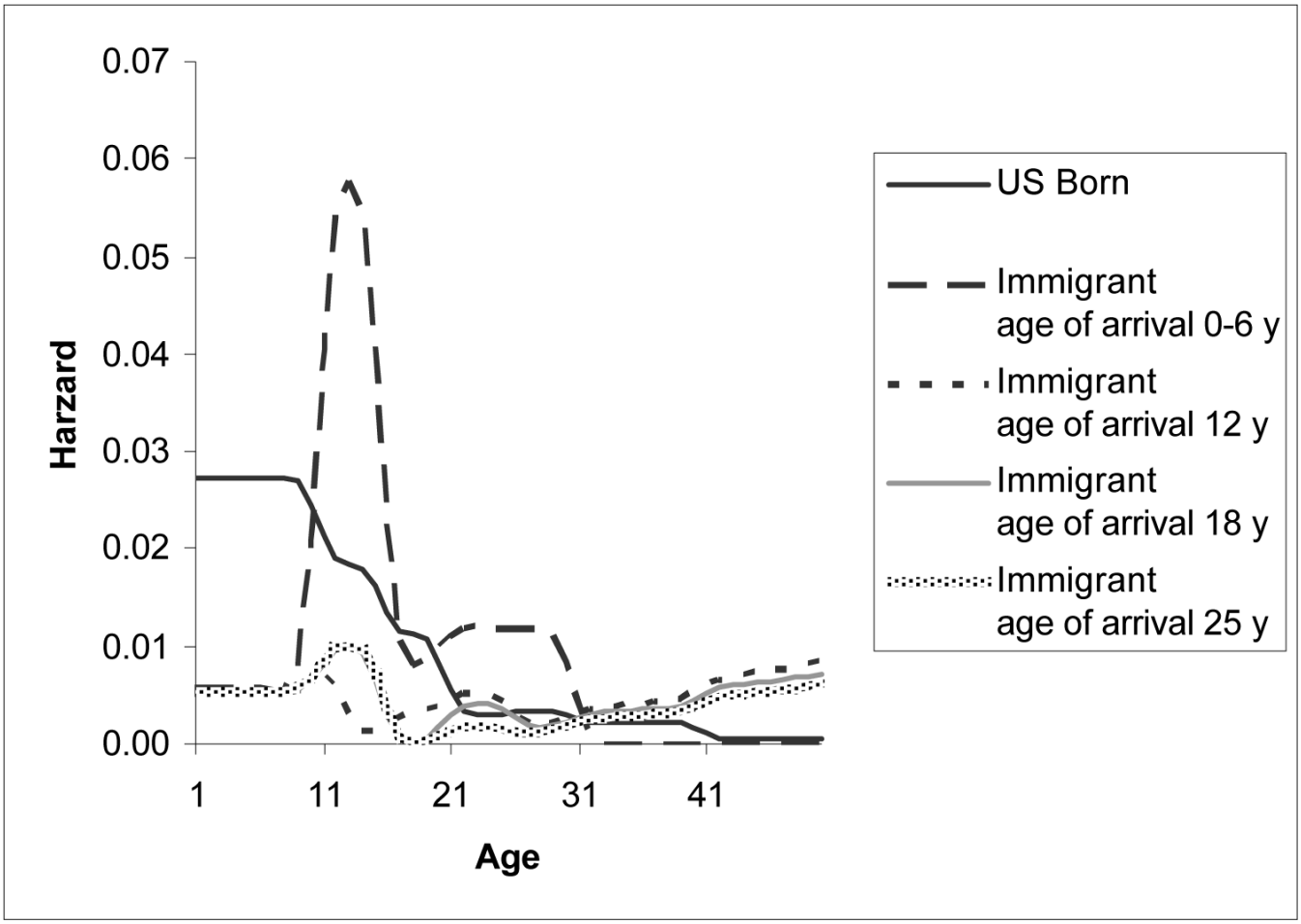

Figure 1.
Weighted exponential survival models fit by age for male depressive disorders by nativity and age of arrival into the U.S.

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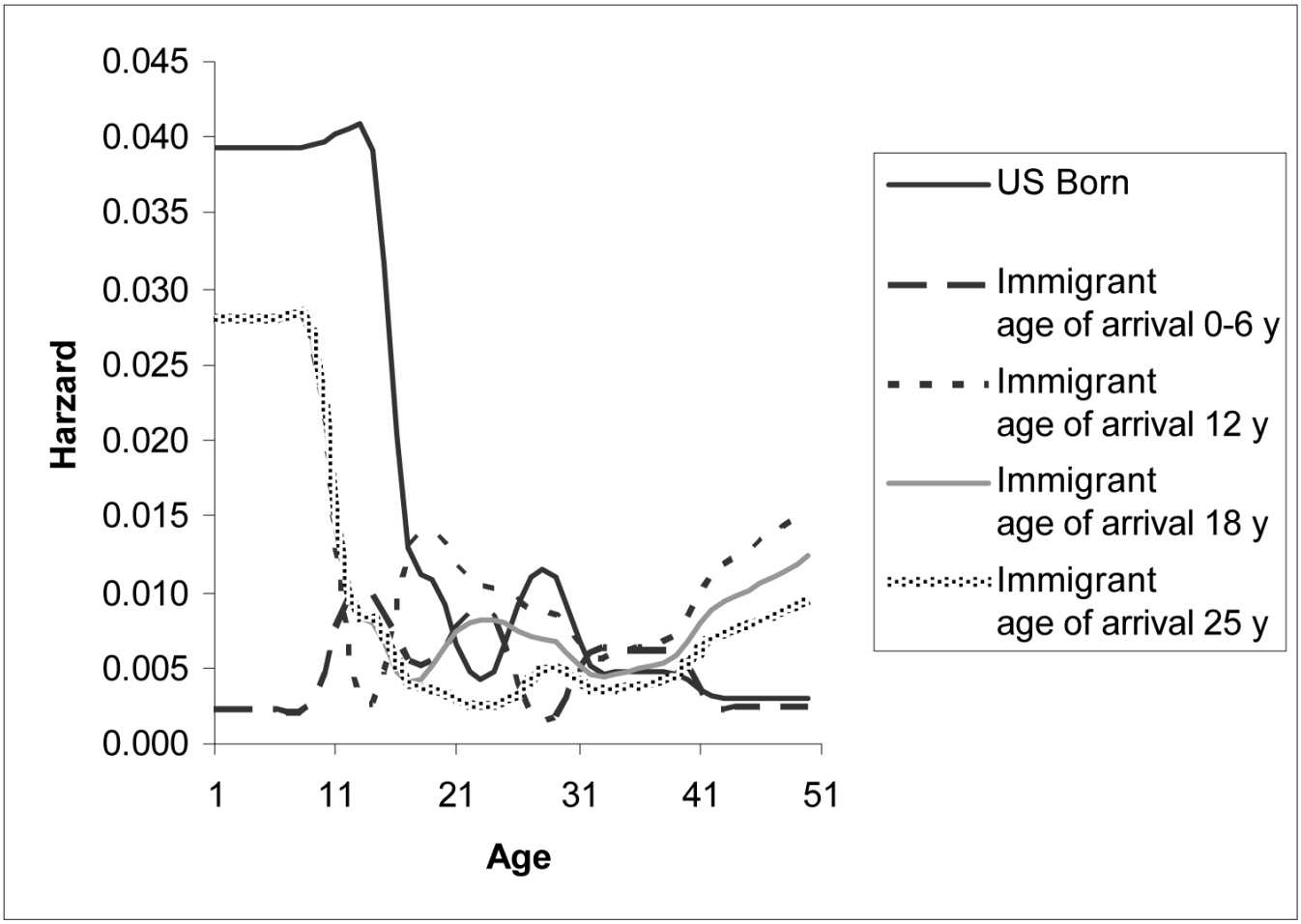

Figure 2.
Weighted exponential survival models fit by age for female depressive disorders by nativity and age of arrival into the U.S.

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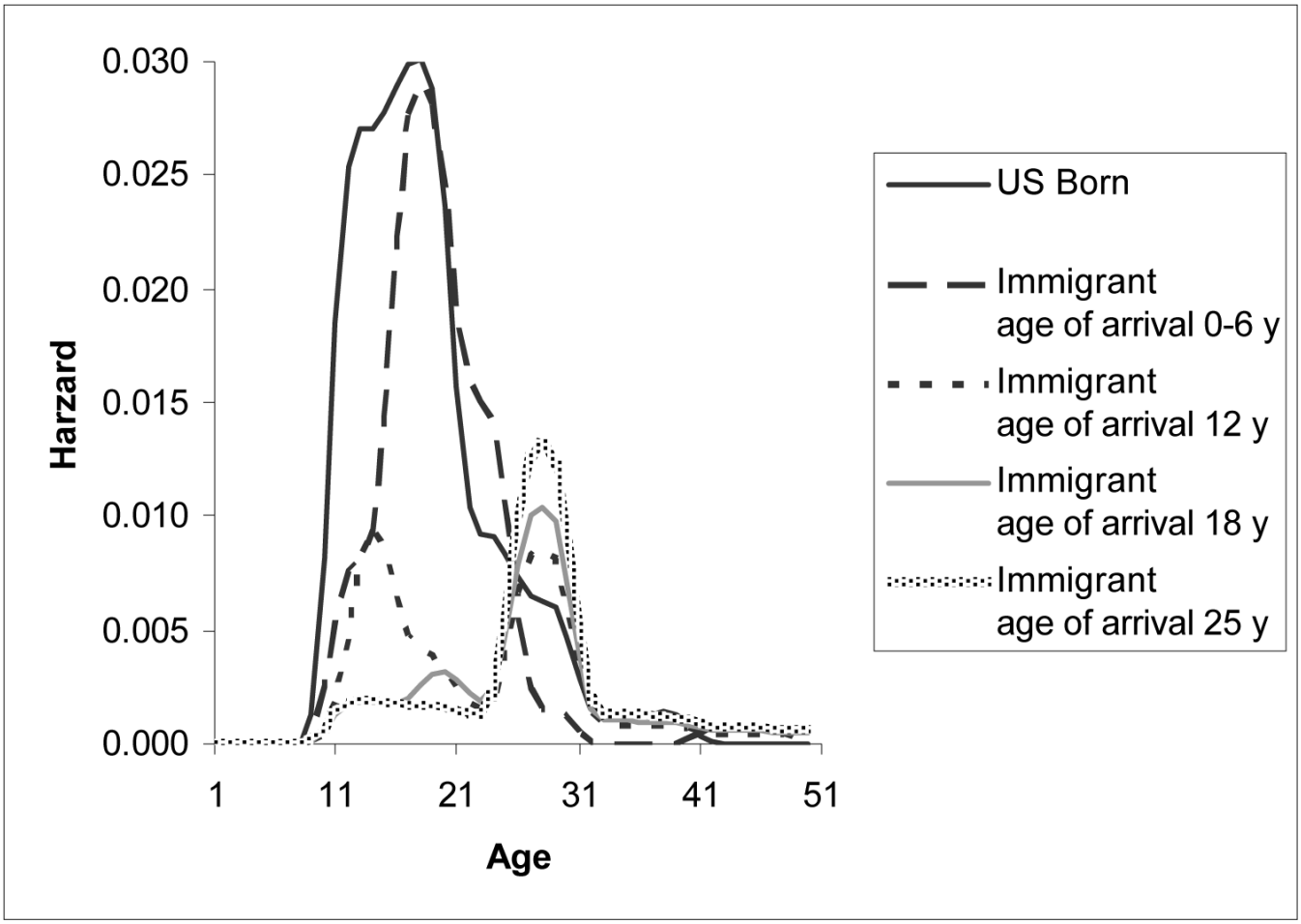

Figure 3.
Weighted exponential survival models fit by age for male substance use disorders by nativity and age of arrival into the U.S.

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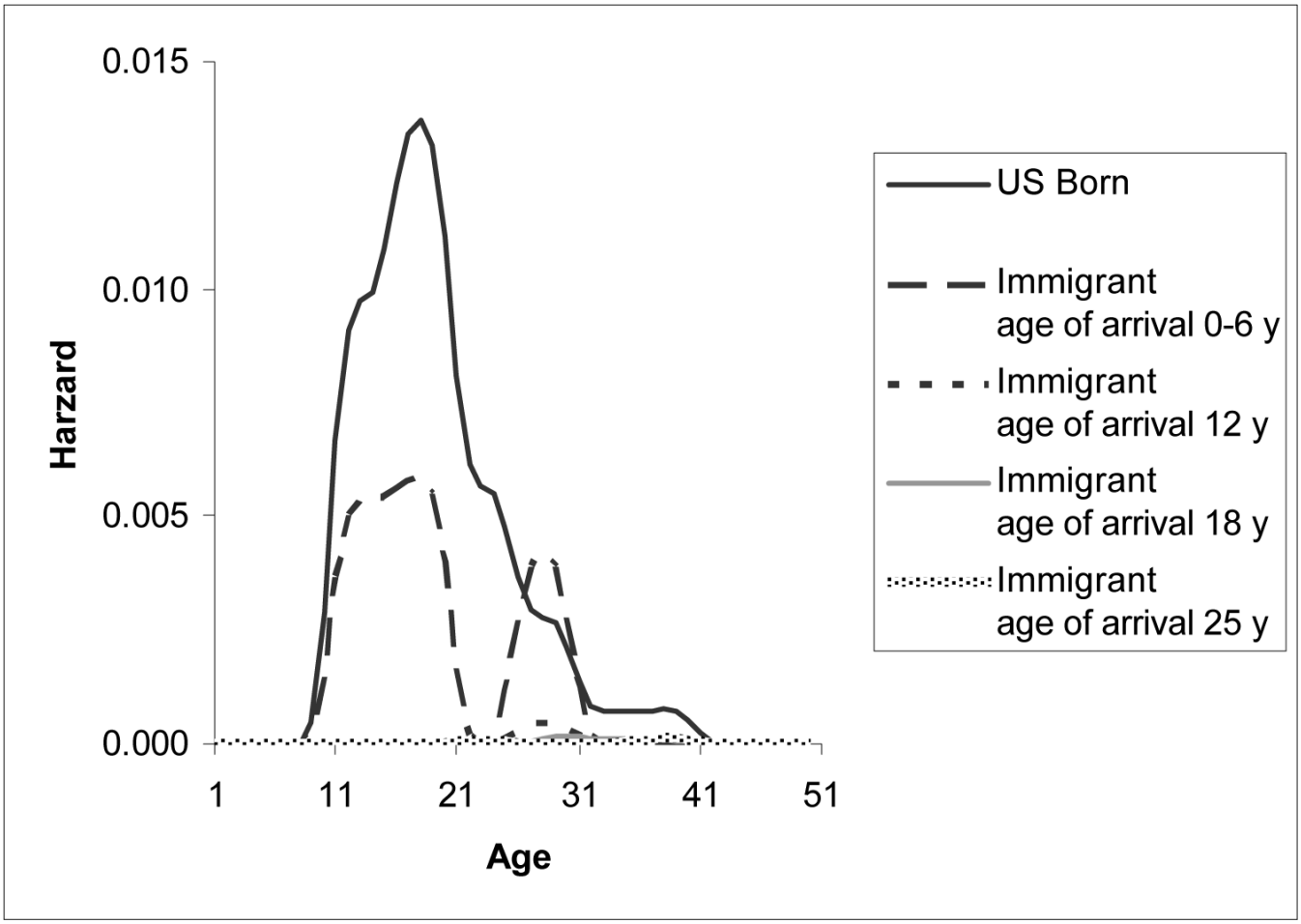

Figure 4.
Weighted exponential survival models fit by age for female substance use disorders by nativity and age of arrival into the U.S.

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Table 1

Characteristics[1] of the NLAAS Latino Sample.

	All persons	U.S. born Latinos	Immigrants			Test of differences among nativity/age of arrivals groups
			Age of arrival 0–6 y	Age of arrival 7–17 y	Age of arrival ≥18 y	
Sample						
N	2546	924	203	378	1041	
weighted %	100.0	41.6	6.8	16.4	35.2	
Sex						NS
Male	51.5	51.4	50.3	57.3	49.2	
Female	48.5	48.6	49.7	42.7	50.8	
Age (y)						***
18–24	20.7	28.6	27.7	30.1	5.8	
25–34	28.3	24.7	32.2	32.3	29.8	
35–49	30.1	27.3	26.8	26.8	35.6	
50–64	13.4	11.6	10.2	9.6	18.0	
65+	7.5	7.8	3.1	1.3	10.8	
Mean age (y)	38.0	36.7	34.3	32.7	42.8	***
Immigrants, time in the U.S.						***[2]
In U.S., 5 years or less	-	-	0.0	12.6	21.6	
In U.S., more than 5 years	-	-	100.0	87.4	78.4	
Immigrants, mean time (y) in the U.S.	18.3[3]	-	30.8	19.4	15.5	***[4]
Birth cohort						***
Born before 1960	32.5	30.4	23.4	19.7	42.9	
Born after 1960	67.5	69.6	76.6	80.3	57.1	
Prevalence of psychiatric disorders						
Any lifetime depressive disorder	15.6	17.4	17.2	14.9	13.5	NS
Any 12-month depressive disorder	8.9	10.1	9.2	8.7	7.5	NS
Any lifetime anxiety disorder	15.4	17.4	19.2	12.7	13.4	NS
Any 12-month anxiety disorder	9.6	10.6	11.0	8.1	8.9	NS
Any lifetime substance-use disorder	11.3	18.9	11.3	5.4	4.9	***

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	All persons	U.S. born Latinos	Immigrants			Test of differences among nativity/age of arrivals groups
			Age of arrival 0–6 y	Age of arrival 7–17 y	Age of arrival ≥18 y	
Any 12 month substance-use disorder	2.9	5.1	2.1	1.6	1.0	**

NS = nonsignificant

[1] All data except sample size and means reported as percentages.

[2] Test of difference among all immigrant subgroups. Test between immigrants age of arrival 7–17 and ≥18 is significant at $p = 0.01$.

[3] Immigrants only.

[4] Test of difference among all immigrant subgroups. Test between immigrants' age of arrival 7–17 and ≥18 is significant at $p < 0.001$.

** $p < 0.01$

*** $p < 0.001$.

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Table 2

Incidence rates for onset of depressive, anxiety, and substance-use disorders for U.S.-born Latinos and for immigrant Latinos before and after arrival to the U.S. by age of arrival [1].

		Immigrants		
	U.S. born	**Age of arrival 0–6 y**	**Age of arrival 7–17 y**	**Age of arrival ≥18 y**
Sample *N*	924	203	378	1041
Any depressive disorder				
Number of onsets observed before arrival in U.S.[2]	-	0	20	92
Number of onsets observed after arrival in U.S.	182	36	50	81
Incidence rate (per 1000 person-year) before arrival in U.S.	-	0.0	3.7	2.8
Incidence rate (per 1000 person-year) after arrival in U.S.	5	5.3	4.7	4.2
Any anxiety disorder				
Number of onsets observed before arrival in U.S.[2]	-	2	29	115
Number of onsets observed after arrival in U.S.	166	36	30	47
Incidence rate (per 1000 person-year) before arrival in U.S.	-	5.0	5.3	3.0
Incidence rate (per 1000 person-year) after arrival in U.S.	5.2	5.8	2.6	2.8
Substance-use disorder				
Number of onsets observed before arrival in U.S.[2]	-	0	2	19
Number of onsets observed after arrival in U.S.	158	21	17	27
Incidence rate (per 1000 person-year) before arrival in U.S.	-	0.0	0.3	0.6
Incidence rate (per 1000 person-year) after arrival in U.S.	5.6	3.8	2.1	2.1

Incidence rates are crude weighted estimates (not age-adjusted).

[1] Sample *N* and numbers of observed onsets are unweighted.

[2] Onsets that occurred in the same year as arrival in U.S. are included in "before" category.

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Table 3

Cox Proportional Hazards Models (Hazard Ratios with 95% Confidence Interval) for Onset of Depressive, Anxiety, and Substance-Use Disorders for Latinos by Sex.

	Depressive disorders		Anxiety disorders		Substance-use disorders	
	Male	**Female**	**Male**	**Female**	**Male**	**Female**
US born	1	1	1	1	1	1
Immigrant, age of arrival 0−6 y	1.36 (0.54−3.42)	0.64 (0.33−1.25)	1.27 (0.60−2.65)	0.88 (0.59−1.31)	0.59 (0.28−1.25)	0.49 (0.20−1.24)
Immigrant, age of arrival ≥7 y						
In country of origin	0.56 (0.32−0.97)*	0.74 (0.52−1.06)	0.55 (0.33−0.94)*	0.65 (0.47−0.91)*	0.15 (0.08−0.28)***	0.01 (0.00−0.06)***
In US, ≤5 y	0.45 (0.17−1.15)	0.54 (0.30−0.97)*	0.74 (0.31−1.79)	0.55 (0.26−1.14)	0.39 (0.22−0.71)**	-[1]
In US, >5 y	0.88 (0.53−1.46)	0.94 (0.64−1.37)	0.66 (0.29−1.54)	1.09 (0.69−1.74)	0.49 (0.30−0.81)**	0.04 (0.01−0.14)***
Birth cohort						
Born before 1960	0.46 (0.30−0.69)***	0.36 (0.26−0.48)***	1.09 (0.72−1.66)	0.78 (0.57−1.07)	0.98 (0.70−1.37)	0.41 (0.23−0.74)**
Born after 1960	1	1	1	1	1	1

[1] No SUD onsets observed for female immigrants in US, ≤5 y.

* $p < 0.05$

** $p < 0.01$

*** $p < 0.001$.